

1.

$$\rho(x) = \begin{cases} -qN_A, & -x_p \leq x \leq 0 \\ +qN_D, & 0 \leq x \leq x_n \\ 0, & |x| > \text{depletion region} \end{cases}$$

Here, q is the electronic charge, N_A is the acceptor concentration (m^{-3}) in the p-region, and N_D is the donor concentration (m^{-3}) in the n-region. The depletion region extends from $-x_p$ to x_n . At equilibrium, mobile carriers are depleted in the region $-x_p \leq x \leq x_n$, leaving behind fixed ionized dopants that give rise to the space charge density $\rho(x)$.

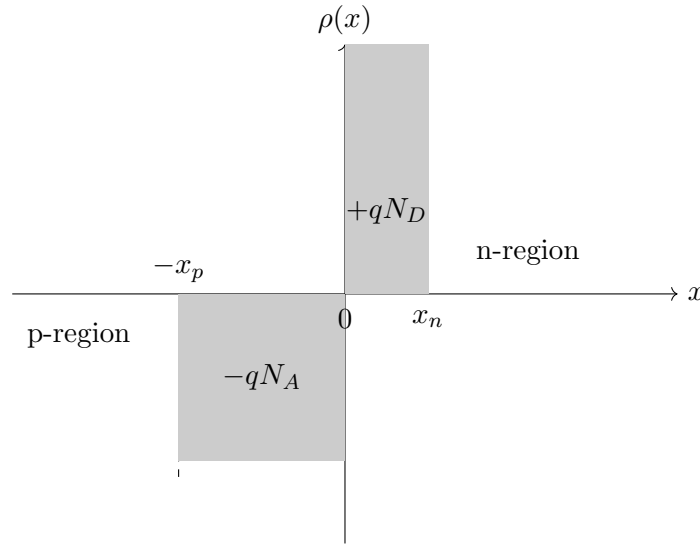


Figure 1: Charge distribution

- Identify what the free charges are in this case and understand what the term "free" truly refers to.
- Find the potential drop across the junction.
- Plot the $\vec{E}(x)$ Vs x .
- Plot the $V(x)$ Vs x .

Solution:

- Free charges: what does "free" mean here?**

In Maxwell's equations, the word *free* does *not* mean "mobile". It means: the charge density is treated as an explicit source term in Gauss/Poisson equation (i.e., it is specified independently, not arising from polarization).

In the depletion region of an abrupt pn junction, mobile carriers (electrons/holes) are depleted, leaving behind *fixed ionized dopants*:

- p-side: negatively charged ionized acceptors $\Rightarrow \rho(x) = -qN_A$
- n-side: positively charged ionized donors $\Rightarrow \rho(x) = +qN_D$

Hence, under depletion approximation,

$$\rho(x) = \begin{cases} -qN_A, & -x_p \leq x \leq 0 \\ +qN_D, & 0 \leq x \leq x_n \\ 0, & |x| > \text{depletion region.} \end{cases}$$

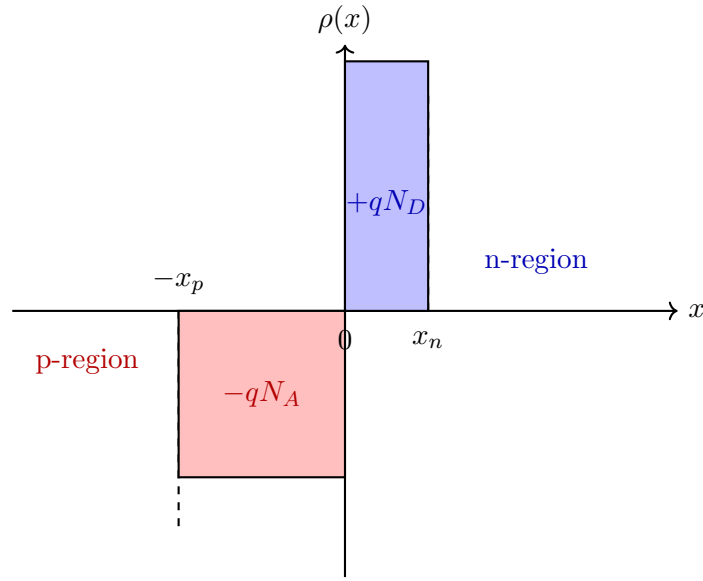


Figure 2: Space-charge density $\rho(x)$ in depletion approximation (colored).

(b) **Potential drop across the junction (built-in potential)**

In 1D, Poisson/Gauss law gives

$$\frac{dE_x(x)}{dx} = \frac{\rho(x)}{\varepsilon_s}, \quad \mathbf{E}(x) = E_x(x) \hat{\mathbf{x}}.$$

Integrate in each region. Using the standard depletion-boundary conditions

$$E_x(-x_p) = 0, \quad E_x(x_n) = 0,$$

we obtain

$$E_x(x) = \begin{cases} -\frac{qN_A}{\varepsilon_s}(x + x_p), & -x_p \leq x \leq 0, \\ \frac{qN_D}{\varepsilon_s}(x - x_n), & 0 \leq x \leq x_n. \end{cases}$$

Note: for $0 \leq x \leq x_n$, $x - x_n \leq 0 \Rightarrow E_x(x) \leq 0$. So E_x does *not* reverse sign within the depletion region.

Continuity of E_x at $x = 0$ yields the depletion charge neutrality:

$$E_x(0^-) = E_x(0^+) \Rightarrow \frac{qN_A x_p}{\varepsilon_s} = \frac{qN_D x_n}{\varepsilon_s} \Rightarrow N_A x_p = N_D x_n.$$

Now use

$$E_x(x) = -\frac{dV(x)}{dx}.$$

Choose the reference $V(-x_p) = 0$. Integrating piecewise (and enforcing continuity at $x = 0$) gives the *correct* potential:

$$V(x) = \begin{cases} \frac{qN_A}{2\varepsilon_s}(x + x_p)^2, & -x_p \leq x \leq 0, \\ \frac{qN_A x_p^2}{2\varepsilon_s} + \frac{qN_D}{\varepsilon_s} \left(x_n x - \frac{x^2}{2} \right), & 0 \leq x \leq x_n. \end{cases}$$

Intuition (sign + shape)

Since $E_x(x) \leq 0$ throughout the depletion region, we have $\frac{dV}{dx} = -E_x \geq 0$, hence $V(x)$ must be *monotonic* across the depletion region and must show a *net* potential drop $V_{bi} \neq 0$.

(c) Plot of $E(x)$ versus x

Below is a clean *shape plot*. It is normalized to show the triangular form: $E_x(-x_p) = 0$, $E_x(0) = -E_{\max}$, $E_x(x_n) = 0$. (Actual scale is $E_{\max} = \frac{qN_A x_p}{\varepsilon_s} = \frac{qN_D x_n}{\varepsilon_s}$.)

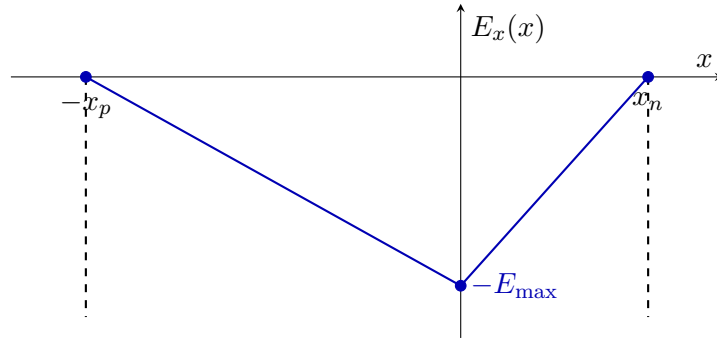


Figure 3: Electric field $E_x(x)$ across the depletion region (sign shown, colored).

(d) Plot of $V(x)$ versus x (correct monotonic behavior)

Using the same normalization and reference $V(-x_p) = 0$, a consistent shape plot is:

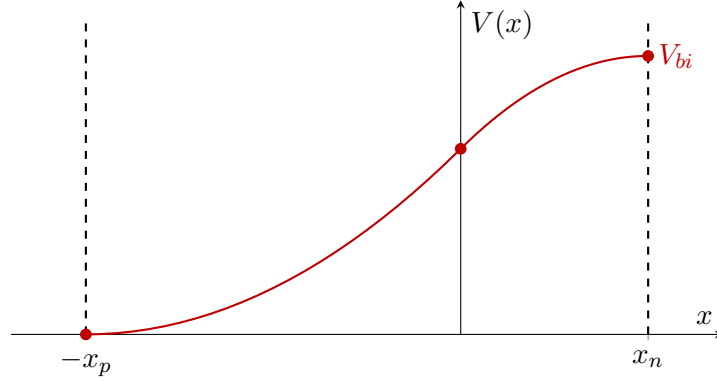


Figure 4: Electrostatic potential $V(x)$ across depletion region (monotonic, net drop $V_{bi} > 0$, colored).

2. Consider the two scalar potential functions

$$V_1(x, y) = x \text{ and } V_2(x, y) = x + e^{-y} \sin x.$$

- Show that both V_1 and V_2 satisfy Laplace's equation.
- Verify that both potentials satisfy the boundary condition $V = 0$ on the boundary $x = 0$.
- Are V_1 and V_2 identical everywhere in space? Justify your answer.
- Does the uniqueness theorem apply to this problem? Explain clearly which assumptions of the theorem are or are not satisfied.

Solution:

We are given the two potential functions

$$V_1(x, y) = x, \quad V_2(x, y) = x + e^{-y} \sin x.$$

(a) Verification of Laplace's equation

Laplace's equation in two dimensions is

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2}.$$

For V_1 ,

$$\frac{\partial^2 V_1}{\partial x^2} = 0, \quad \frac{\partial^2 V_1}{\partial y^2} = 0,$$

hence,

$$\nabla^2 V_1 = 0.$$

For V_2 ,

$$\frac{\partial^2}{\partial x^2}(e^{-y} \sin x) = -e^{-y} \sin x,$$

$$\frac{\partial^2}{\partial y^2}(e^{-y} \sin x) = e^{-y} \sin x.$$

Therefore,

$$\nabla^2 V_2 = -e^{-y} \sin x + e^{-y} \sin x = 0.$$

Thus, both V_1 and V_2 satisfy Laplace's equation.

(b) **Verification of boundary condition**

On the boundary $x = 0$,

$$V_1(0, y) = 0,$$

$$V_2(0, y) = 0 + e^{-y} \sin 0 = 0.$$

Hence both potentials satisfy the boundary condition

$$V = 0 \quad \text{on } x = 0.$$

(c) **Are V_1 and V_2 identical everywhere?**

The difference between the two potentials is

$$V_2 - V_1 = e^{-y} \sin x,$$

which is not identically zero.

Therefore,

$$V_1 \neq V_2$$

in general, even though both satisfy Laplace's equation and the same boundary condition on $x = 0$.

(d) **Applicability of the uniqueness theorem**

The uniqueness theorem for Laplace's equation requires that the potential be specified on a *closed boundary*, or that sufficient boundary conditions be provided to uniquely determine the solution.

In this problem, the potential is specified only on the line $x = 0$, which does not form a closed boundary. No conditions are imposed as $x \rightarrow \infty$ or $y \rightarrow \pm\infty$.

Therefore, the boundary conditions are incomplete and the uniqueness theorem does not apply. Hence, multiple solutions satisfying the given conditions are possible.

3. Two concentric conducting spheres of radii $a = 5$ mm and $b = 20$ mm are separated by air. The inner sphere is maintained at a potential of 100 V, while the outer sphere is grounded.
- Assuming electrostatic equilibrium, determine the electric potential $V(r)$ in the region $a \leq r \leq b$ by solving Laplace's equation, and verify the result by comparison with the electric field obtained from Gauss's law.
 - Obtain an expression for the radial electric field $E_r(r)$ in the region between the spheres and determine the location and value of the maximum electric field.
 - Find the radius at which the electric potential is 20 V.
 - Plot the electric potential and electric field in a two-dimensional circular cross-section of the region between the spheres, clearly indicating the equipotential surfaces and field direction.

Solution:

- (a) **Electric potential $V(r)$ in the region $a \leq r \leq b$**

Due to spherical symmetry and electrostatic equilibrium, the potential depends only on the radial coordinate r .

In the charge-free region between the conductors, Laplace's equation in spherical coordinates reduces to

$$\nabla^2 V = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV}{dr} \right) = 0.$$

Integrating once,

$$r^2 \frac{dV}{dr} = C_1,$$

$$\frac{dV}{dr} = \frac{C_1}{r^2}.$$

Integrating again,

$$V(r) = A + \frac{B}{r}.$$

The boundary conditions are

$$V(a) = 100 \text{ V}, \quad V(b) = 0.$$

Applying them,

$$A + \frac{B}{a} = 100, \quad A + \frac{B}{b} = 0.$$

Solving,

$$B = \frac{100ab}{b-a}, \quad A = -\frac{100a}{b-a}.$$

Therefore,

$$V(r) = 100 \frac{\frac{1}{r} - \frac{1}{b}}{\frac{1}{a} - \frac{1}{b}}$$

Verification using Gauss's Law:

For a spherical Gaussian surface of radius r ,

$$\oint \mathbf{E} \cdot d\mathbf{A} = E_r(4\pi r^2) = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

Hence,

$$E_r(r) = \frac{Q}{4\pi\varepsilon_0 r^2}.$$

Since $E_r = -dV/dr$, the same $1/r^2$ dependence confirms the solution.

(b) Radial electric field and maximum value

The electric field is

$$E_r(r) = -\frac{dV}{dr}.$$

Differentiating,

$$E_r(r) = \frac{100}{\left(\frac{1}{a} - \frac{1}{b}\right)} \frac{1}{r^2}$$

The magnitude increases as r decreases. Therefore, the maximum electric field occurs at

$$r = a.$$

$$E_{\text{max}} = \frac{100}{\left(\frac{1}{a} - \frac{1}{b}\right)} \frac{1}{a^2}$$

(c) **Radius at which $V = 20$ V**

Set

$$V(r) = 20 \text{ V.}$$

$$20 = 100 \frac{\frac{1}{r} - \frac{1}{b}}{\frac{1}{a} - \frac{1}{b}}$$

Solving for r ,

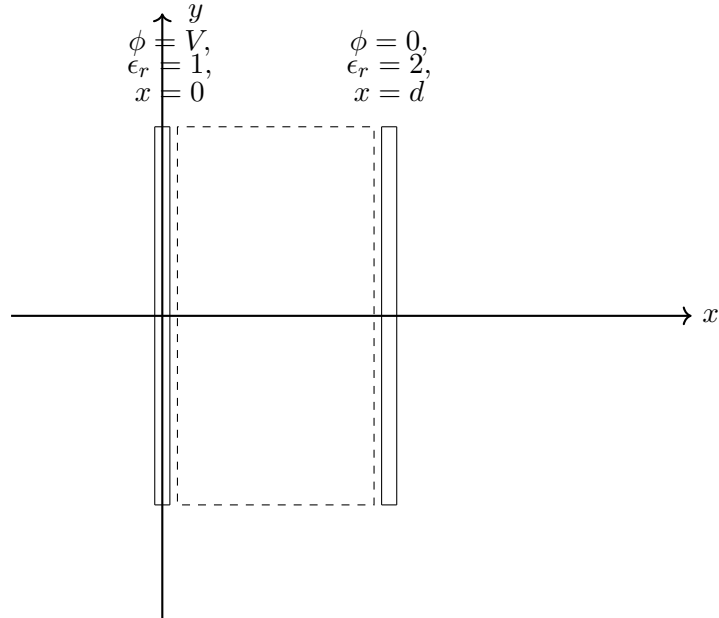
$$r = 12.5 \text{ mm}$$

(d) **Potential and field distribution (qualitative plot description)**

In a two-dimensional circular cross-section:

- Equipotential surfaces are concentric circles.
- Electric field lines are radial.
- Equipotential spacing decreases near the inner sphere, indicating higher electric field magnitude.
- The field direction is radially outward (from the 100 V sphere toward the grounded sphere).

4. The space between the plates of a parallel-plate capacitor is filled with dielectric material whose dielectric constant varies linearly from 1 at the leftmost plate ($x = 0$) to 2 at the rightmost plate ($x = d$). The capacitor is connected to a battery of voltage V . Find all the bound charge, and check that the total is zero.



Solution:

The relative permittivity varies linearly with x :

$$\epsilon_r(x) = 1 + \frac{x}{d}$$

The potential difference between the plates is maintained constant:

$$V = \int_0^d E(x) dx$$

Since there is no free charge in the dielectric region,

$$\nabla \cdot \mathbf{D} = \rho_{\text{free}} = 0$$

Hence the displacement field $\mathbf{D}$ is constant.

Using

$$\mathbf{D} = \epsilon_0 \epsilon_r(x) \mathbf{E}(x),$$

we obtain

$$V = \int_0^d \frac{D}{\epsilon_0 \epsilon_r(x)} dx = \int_0^d \frac{D}{\epsilon_0 \left(1 + \frac{x}{d}\right)} dx.$$

$$V = \frac{D}{\epsilon_0} d \ln(2)$$

$$D = \frac{\epsilon_0 V}{d \ln(2)}$$

Therefore,

$$E(x) = \frac{D}{\epsilon_0 \left(1 + \frac{x}{d}\right)} = \left[\frac{V}{(d+x) \ln(2)} \right] \hat{i}$$

Polarization

Using (from Griffiths),

$$\mathbf{P}(x) = \epsilon_0 [\epsilon_r(x) - 1] \mathbf{E}(x),$$

$$\mathbf{P}(x) = \epsilon_0 \frac{xV}{d(d+x) \ln(2)} \hat{i}.$$

Bound surface charge

$$\sigma_b = \mathbf{P} \cdot \hat{n}$$

At $x = 0$:

$$P(0) = 0 \quad \Rightarrow \quad q_b(0) = 0.$$

At $x = d$:

$$\sigma_b(d) = \frac{\epsilon_0 V}{2d \ln(2)}.$$

Total surface bound charge at $x = d$:

$$q_{b,\text{surface}} = \frac{\epsilon_0 V A}{2d \ln(2)}$$

Bound volume charge

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{dP}{dx}.$$

Differentiating,

$$\rho_b(x) = -\frac{\epsilon_0 V}{(d+x)^2 \ln(2)}$$

Total bound charge in the volume:

$$\begin{aligned} q_{b,\text{volume}} &= \int_A \int_0^d \rho_b(x) dx dA \\ &= A \int_0^d -\frac{\epsilon_0 V}{(d+x)^2 \ln(2)} dx \\ &= \frac{\epsilon_0 V A}{(d+x) \ln(2)} \Big|_0^d = -\frac{\epsilon_0 V A}{2d \ln(2)}. \end{aligned}$$

Total bound charge

$$q_{b,\text{net}} = q_{b,\text{surface}} + q_{b,\text{volume}}$$

$$= \frac{\epsilon_0 V A}{2d \ln(2)} - \frac{\epsilon_0 V A}{2d \ln(2)}$$

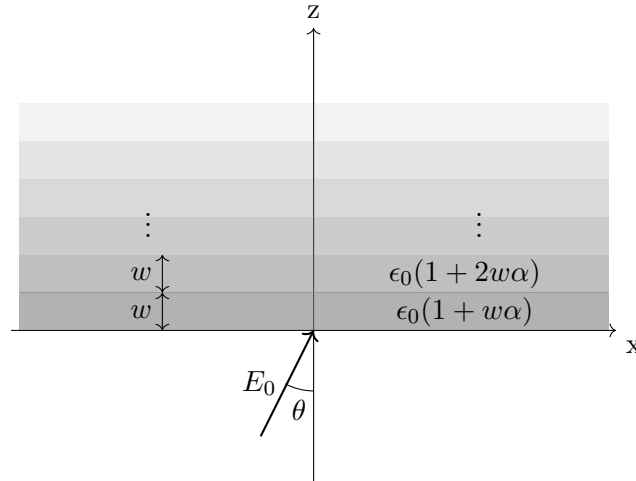
$$\boxed{q_{b,\text{net}} = 0}$$

Hence, the total bound charge in the system is zero.

5. Consider a two-dimensional system in the x-z plane where the region $z > 0$ is filled with horizontal dielectric slabs of width w . The i-th slab has the permittivity

$$\epsilon_i = \epsilon_0(1 + \alpha i w), \quad \alpha > 0$$

An electric field enters the region from $z < 0$, making an angle θ with the z-axis.



- (a) Using the standard boundary conditions at each interface between layers, explain qualitatively how the electric field bends at each layer boundary.
- (b) Sketch a field line passing through several layers, showing how it changes direction at each interface.
- (c) Consider the case where $w \rightarrow 0$. Explain how the field trajectory changes in this limit and sketch a field line.

Solution:

(a) **Bending of the electric field at each interface**

At the interface between two dielectric media (with no free surface charge), the boundary conditions are

$$E_{1t} = E_{2t}$$

$$\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

The tangential component remains continuous across every interface.

If the incident field makes an angle θ with the z -axis, then

$$E_{0t} = E_0 \sin \theta, \quad E_{0n} = E_0 \cos \theta.$$

Since the slabs are horizontal, the tangential component is along x and remains constant through all layers:

$$E_{it} = E_{0t}.$$

For the normal component:

At the first interface,

$$\epsilon_0 E_{0n} = \epsilon_0(1 + w\alpha)E_{1n}$$

$$\Rightarrow E_{1n} = \frac{E_{0n}}{1 + w\alpha}.$$

At the second interface,

$$\epsilon_0(1 + w\alpha)E_{1n} = \epsilon_0(1 + 2w\alpha)E_{2n}$$

$$\Rightarrow E_{2n} = \frac{E_{0n}}{1 + 2w\alpha}.$$

Similarly, in the i -th slab,

$$E_{in} = \frac{E_{0n}}{1 + iw\alpha}$$

Thus, as we move upward through the layers:

- The tangential component remains constant.
- The normal component decreases.

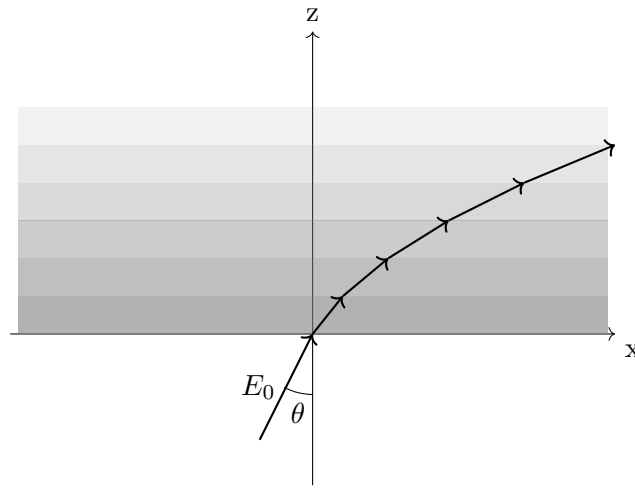
Therefore, the electric field bends progressively toward the positive x -direction. The angle in the i -th slab is

$$\tan \theta_i = \frac{E_{it}}{E_{in}} = (1 + iw\alpha) \tan \theta.$$

Hence, the angle increases as the permittivity increases.

(b) **Sketch of the field line**

A field line passing through multiple layers changes direction at each interface. The slope increases gradually because the normal component decreases while the tangential component remains constant.



(c) **Limit as $w \rightarrow 0$**

As $w \rightarrow 0$, the medium becomes continuously graded in permittivity:

$$\epsilon(z) = \epsilon_0(1 + \alpha z).$$

The normal component becomes

$$E_n(z) = \frac{E_{0n}}{1 + \alpha z}$$

The tangential component remains constant.

Hence,

$$\tan \theta(z) = \frac{E_t}{E_n(z)} = (1 + \alpha z) \tan \theta.$$

Thus, the angle increases smoothly with z .

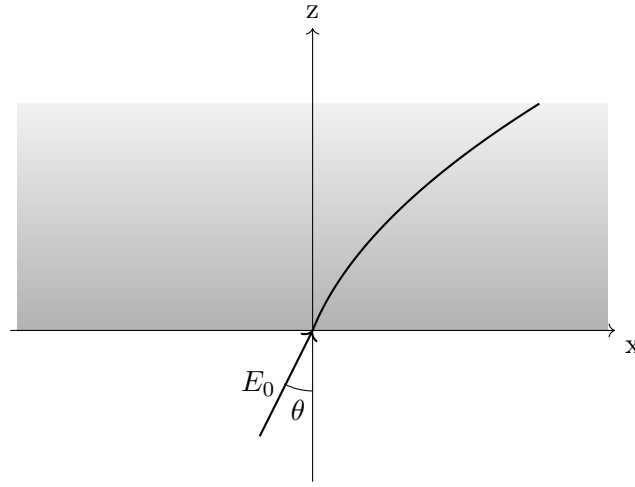
The trajectory of a field line satisfies

$$\frac{dx}{dz} = \tan \theta(z) = (1 + \alpha z) \tan \theta.$$

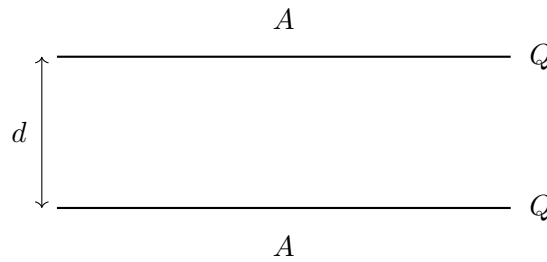
Integrating,

$$x(z) = \tan \theta \left(z + \frac{\alpha z^2}{2} \right).$$

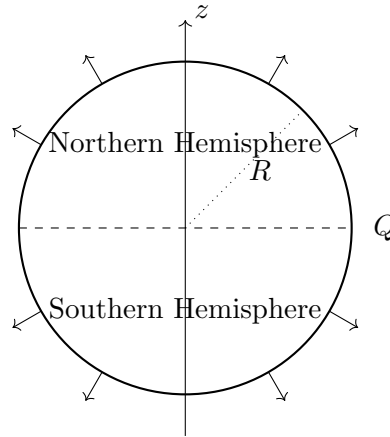
Hence, the field line follows a **parabolic path** in the continuously graded dielectric.



6. (a) Two large parallel metal plates, each of area A , are held a small distance d apart. Each plate is given a charge Q . Assuming edge effects are negligible, find the **electrostatic pressure** exerted on the plates and hence determine the force acting on each plate.



- (b) A conducting metal sphere of radius R carries a total charge Q . What is the force of repulsion between the **northern hemisphere** and the **southern hemisphere** of the sphere?



Solution:

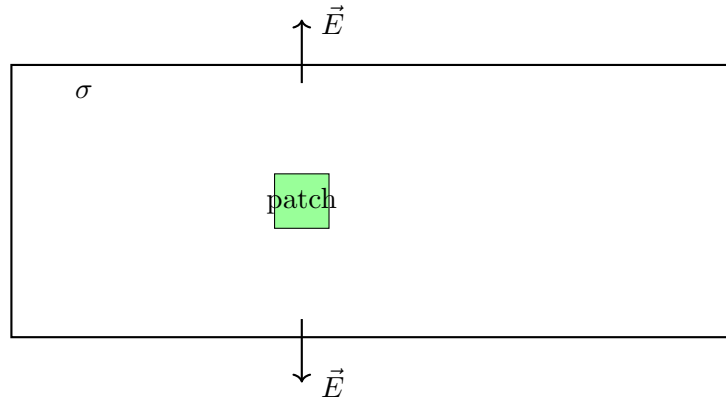
(a) **Electrostatic pressure on two charged plates**

Consider an infinite plane sheet carrying uniform surface charge density σ .

From Gauss's law, the electric field due to a single infinite sheet is

$$E = \frac{\sigma}{2\epsilon_0}$$

on each side of the sheet, directed normally outward.



To find the force on a small patch of area ΔA , note that a surface element does not act on itself. Therefore, the force on the patch is due to the field created by the rest of the sheet:

$$E_{\text{rest}} = \frac{\sigma}{2\epsilon_0}.$$

Charge on the patch:

$$\Delta q = \sigma \Delta A.$$

Force on the patch:

$$\Delta F = \Delta q E_{\text{rest}} = \sigma \Delta A \cdot \frac{\sigma}{2\varepsilon_0}.$$

Hence the electrostatic pressure is

$$P = \frac{\Delta F}{\Delta A} = \boxed{\frac{\sigma^2}{2\varepsilon_0}}.$$

Using $\sigma = \frac{Q}{A}$,

$$\boxed{P = \frac{Q^2}{2\varepsilon_0 A^2}}.$$

The force on each plate is

$$\boxed{F = PA = \frac{Q^2}{2\varepsilon_0 A}}.$$

Alternatively, since for a conductor

$$E = \frac{\sigma}{\varepsilon_0},$$

the pressure may also be written as

$$\boxed{P = \frac{1}{2}\varepsilon_0 E^2}.$$

(b) Force between the northern and southern hemispheres of a charged sphere

For a conducting sphere of radius R carrying charge Q , the electric field just outside is

$$E = \frac{1}{4\pi\varepsilon_0} \frac{Q}{R^2}.$$

The surface charge density is

$$\sigma = \frac{Q}{4\pi R^2}.$$

The electrostatic pressure on the surface is

$$P = \frac{\sigma^2}{2\epsilon_0} = \frac{1}{2\epsilon_0} \left(\frac{Q}{4\pi R^2} \right)^2 = \frac{Q^2}{32\pi^2\epsilon_0 R^4}.$$

To find the force separating the hemispheres, integrate the normal component of pressure over the hemisphere.

The projection of a hemisphere onto a plane perpendicular to the force is a circle of area πR^2 . Hence,

$$F = P \times \pi R^2.$$

Substituting,

$$F = \frac{Q^2}{32\pi^2\epsilon_0 R^4} \cdot \pi R^2.$$

$$F = \frac{Q^2}{32\pi\epsilon_0 R^2}.$$

7. A coaxial cable consists of an inner conducting cylinder of radius a and an outer conducting cylinder of radius b . The space between them is filled with two dielectric layers:

- Region 1: $a < r < c$ with permittivity ϵ_1
- Region 2: $c < r < b$ with permittivity ϵ_2

The inner conductor carries charge per unit length $+\lambda$, and the outer conductor carries $-\lambda$.

- (a) Find the electric field in both dielectric regions.
- (b) Determine the potential difference between the conductors.
- (c) Obtain the capacitance per unit length of this structure.
- (d) Find the electrostatic energy stored per unit length.
- (e) Determine the bound surface charge density at the interface $r = c$.

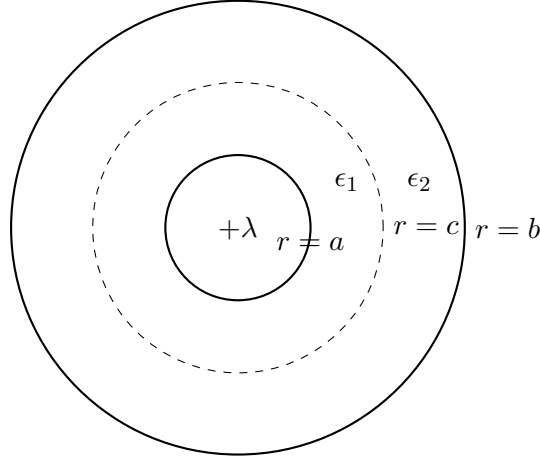


Figure 5: Coaxial capacitor with two dielectric layers

Solution:

(a) **Electric field in both dielectric regions**

Apply Gauss's law in terms of the electric flux density $\mathbf{D}$:

$$\oint \mathbf{D} \cdot d\mathbf{A} = Q_{\text{free, enc.}}$$

Choose a cylindrical Gaussian surface of radius r and length L .

$$D_r(2\pi rL) = \lambda L.$$

$$D_r = \frac{\lambda}{2\pi r}$$

Using $\mathbf{D} = \epsilon\mathbf{E}$:

For region 1 ($a < r < c$),

$$E_1(r) = \frac{\lambda}{2\pi\epsilon_1 r}$$

For region 2 ($c < r < b$),

$$E_2(r) = \frac{\lambda}{2\pi\epsilon_2 r}$$

(b) **Potential difference between the conductors**

The potential difference is

$$V = V(a) - V(b) = - \int_a^b \mathbf{E} \cdot d\mathbf{l}.$$

Since the field is radial,

$$V = - \left(\int_a^c E_1(r) dr + \int_c^b E_2(r) dr \right).$$

$$V = - \left[\frac{\lambda}{2\pi\epsilon_1} \int_a^c \frac{dr}{r} + \frac{\lambda}{2\pi\epsilon_2} \int_c^b \frac{dr}{r} \right].$$

$$V = \frac{\lambda}{2\pi} \left(\frac{1}{\epsilon_1} \ln \frac{c}{a} + \frac{1}{\epsilon_2} \ln \frac{b}{c} \right)$$

(c) **Capacitance per unit length**

By definition,

$$C' = \frac{\lambda}{V}.$$

$$C' = \frac{2\pi}{\frac{1}{\epsilon_1} \ln \frac{c}{a} + \frac{1}{\epsilon_2} \ln \frac{b}{c}}$$

(d) **Electrostatic energy stored per unit length**

Using

$$U' = \frac{1}{2} C' V^2,$$

we obtain the energy per unit length.

Alternatively, using energy density

$$u = \frac{1}{2} \epsilon E^2,$$

$$U' = \frac{1}{2} \int_a^c \epsilon_1 E_1^2 (2\pi r dr) + \frac{1}{2} \int_c^b \epsilon_2 E_2^2 (2\pi r dr),$$

which yields the same result.

(e) **Bound surface charge density at $r = c$**

In a linear dielectric,

$$\mathbf{P} = \epsilon_0(\epsilon_r - 1)\mathbf{E}.$$

At the interface, the bound surface charge density is

$$\sigma_b = \mathbf{P}_2 \cdot \hat{n} - \mathbf{P}_1 \cdot \hat{n}.$$

Since $\hat{n}$ is radial,

$$\mathbf{P}_1 = \epsilon_0(\epsilon_{r1} - 1)\frac{\lambda}{2\pi\epsilon_1 c},$$

$$\mathbf{P}_2 = \epsilon_0(\epsilon_{r2} - 1)\frac{\lambda}{2\pi\epsilon_2 c}.$$

Therefore,

$$\sigma_b = \epsilon_0 \left[(\epsilon_{r2} - 1)\frac{\lambda}{2\pi\epsilon_2 c} - (\epsilon_{r1} - 1)\frac{\lambda}{2\pi\epsilon_1 c} \right]$$